

Quantifying ancestry sorting and its predictability across replicate *Formica* hybrid populations

Supplementary Methods · *Formica polyctena* × *F. aquilonia* hybrid populations

We study 20 independent hybrid populations between *F. polyctena* and *F. aquilonia* to ask whether parental variation sorts *predictably* — the same parental allele fixing across populations, as expected if one parental combination is favoured (directional selection; a Dobzhansky–Muller incompatibility, DMI, with unequal fitness) — or in *random directions* across populations, as expected under drift or under incompatibilities whose alternative combinations are equally fit. Below we define per-locus sorting statistics on ancestry-diagnostic markers, characterise the genomic architecture of differentiation that could confound them, and specify the neutral simulation null under construction. All analyses are reproduced by `dev/R/supplementary_analyses.R`.

S1. Data

Genotypes are unphased diploid dosages (0/1/2) for 1,114,340 SNPs (minor-allele frequency > 0.05) in 194 individuals: 20 hybrid populations (3–20 diploids each) plus allopatric parental references for *F. aquilonia* (n = 15) and *F. polyctena* (n = 15). Each SNP carries a DIEM DiagnosticIndex (DI; higher = more ancestry-diagnostic), a local recombination rate interpolated from the *F. rufa* linkage map (cM/Mb), and membership in an LD-based effective-marker-locus group (eMLG, "cluster") from a two-stage complexity reduction that defines approximately independent units.

S2. Per-locus sorting statistics

For each locus we orient dosage into an *aquilonia-allele frequency* using the sign of the parental frequency difference, then classify each hybrid population as near-fixed toward aquilonia (frequency $\geq 1 - \text{fix_th}$), near-fixed toward polyctena ($\leq \text{fix_th}$), or unsorted ($\text{fix_th} = 0.1$). Writing n_{aqu} , n_{pol} for the counts of near-fixed populations (n_{obs} with data), the amount of sorting decomposes exactly into a directional and a bidirectional part on a single population-fraction scale:

$$\text{prop_fixed} = |\text{uni_score}| + \text{bi_score}, \quad \text{uni_score} = (n_{\text{aqu}} - n_{\text{pol}}) / n_{\text{obs}}, \quad \text{bi_score} = 2 \cdot \min(n_{\text{aqu}}, n_{\text{pol}}) / n_{\text{obs}}.$$

Because all three quantities are fractions of the population panel, a single threshold `sort_th` classifies every unit — SNP or eMLG — into *unsorted*, *unidirectional-aquilonia*, *unidirectional-polyctena*, or *bidirectional*, and genome proportions are weighted tallies of that class. This makes SNP-level, eMLG-level and genome-fraction statements directly comparable (`fix_th` is held constant as a convention; `sort_th` is the cross-unit knob).

Parallelism test. Predictability of direction is tested per locus among the near-fixed populations as $n_{\text{aqu}} \sim \text{Binomial}(n_{\text{fixed}}, 0.5)$, two-sided; rejection indicates a consistent (parallel) direction. With 20 replicate populations this is the per-locus, higher-powered analogue of the cross-population correlation used for three populations by Nouhaud et al. (2022).

Differentiation gate. Near-fixation is evidence of *sorting* only if the locus was segregating in the founding admixture, i.e. the parents differ; otherwise a near-monomorphic locus is trivially "fixed" in every population and mimics perfect parallelism. Loci are therefore gated on DI (or, equivalently here, on parental differentiation): at $\text{DI} > -15$, 100% of loci have parental allele-frequency difference ≥ 0.5

(medians 0.93, 0.90, 0.89, 0.83 at $DI > -15, -20, -25, -30$), so DI and parental differentiation agree across the working range.

Result. At `sort_th` = 0.5 over 16,077 diagnostic ($DI > -15$) SNPs: 89.9% unsorted, 6.9% aquilonia, 2.7% polycтена, 0.4% bidirectional. The aquilonia>polycтена excess is the directional-sorting signal.

Null choice. A per-locus "pooled" null (setting the expected fixation probability to each locus's own mean ancestry) is *circular*: its null probability correlates 0.96 with the observed outcome and yields 0 significant loci, versus ~8,000 under the symmetric null. Because the genome-wide mean aquilonia ancestry at diagnostic loci is 0.50, the symmetric (0.5) null is the appropriate, non-circular choice.

S3. Genomic architecture of differentiation (Tier 1)

We asked whether DI is confounded with recombination — the standard driver of "genomic islands." Genome-wide (Spearman): DI vs recombination $\rho = -0.03$; DI vs within-species π $\rho = -0.46$; DI vs absolute divergence d_{xy} $\rho = +0.13$; DI vs cluster size $\rho = +0.23$.

recomb. decile	cM/Mb	DI	F_{ST}	d_{xy}	π (within)	cluster size
1 (lowest)	0.3	-50.9	0.310	0.423	0.188	1610
5	18.5	-50.4	0.193	0.359	0.215	28
10 (highest)	46.4	-50.3	0.192	0.357	0.214	22

Two conclusions (Fig. S1). First, **DI is essentially orthogonal to recombination** (flat across deciles); its dominant correlate is *reduced within-species diversity* (partly definitional, since a diagnostic marker is near-fixed within each species) rather than elevated absolute divergence. Read through the Cruickshank & Hahn (2014) framework, DI behaves as a relative-differentiation signal, not a barrier-divergence signal. Second, a genuine island signature (elevated F_{ST} and d_{xy} , reduced π , very large clusters) is confined to the lowest recombination decile and is largely orthogonal to DI . Statistically, $DI \perp$ recombination means the two can be treated as **separable predictors of sorting**, reducing confounding downstream. Cluster size is a **collider** of recombination and diversity and is not used as a predictor. (Caveat: π and d_{xy} are computed at MAF-ascertained SNPs, not windowed over all sites, so magnitudes are proxies; the $DI \perp$ recombination result is robust to this.)

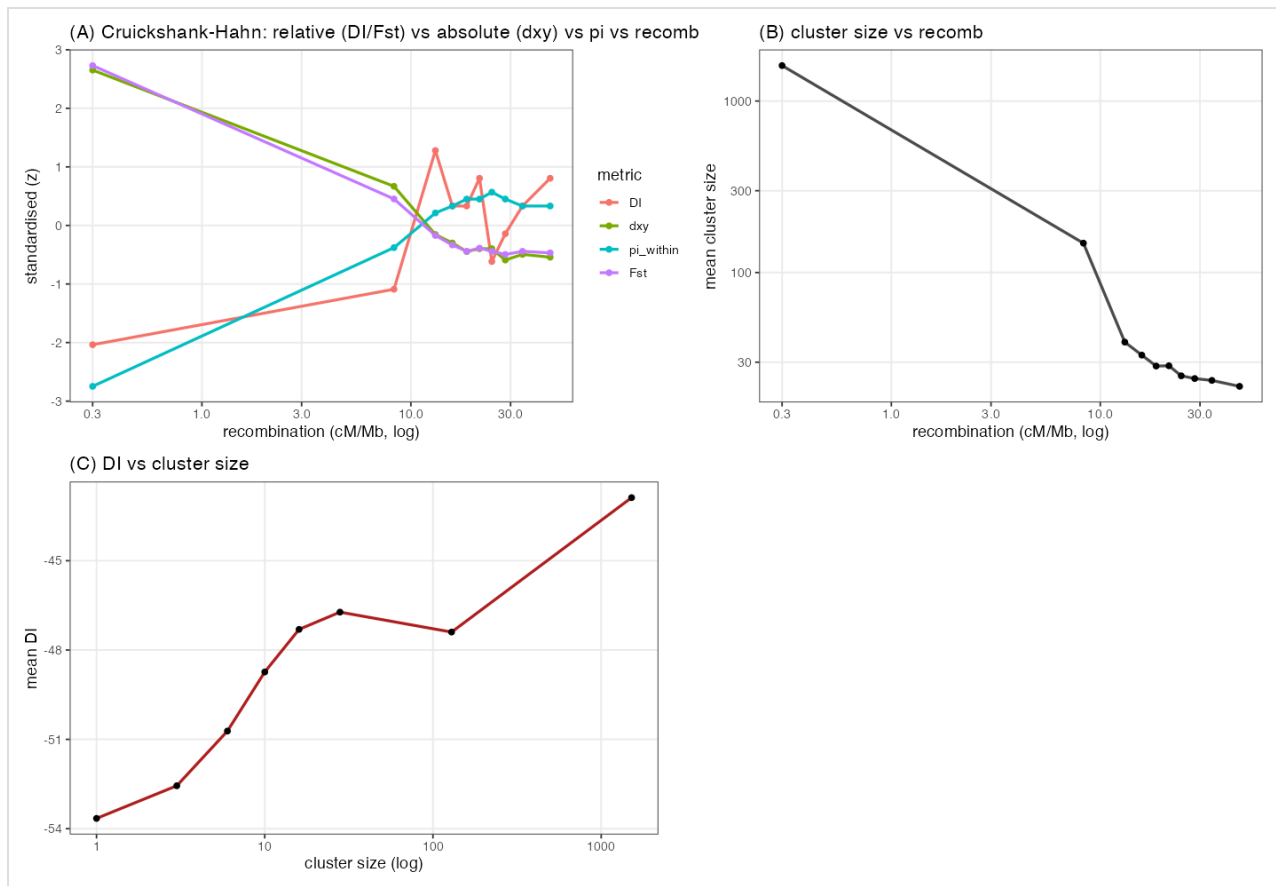


Figure S1. Genomic architecture. (A) Standardised relative (DI, F_{ST}) versus absolute (d_{xy}) differentiation and within-species π across recombination deciles — only the lowest decile shows the island pattern; DI is flat. (B) Cluster size rises steeply only at the very lowest recombination. (C) DI increases modestly with cluster size.

S4. Sorting versus recombination at the independent-unit level (Tier 2)

Neutrally, drift per *independent unit* is recombination-independent; a raw SNP-level relationship between sorting and recombination is inflated by spatial redundancy (linked SNPs in a low-recombination block report a single stochastic event many times). We therefore analysed sorting at the level of LD-pruning representatives (one tag SNP per eMLG cluster), using map-based recombination as the exogenous axis and DI as a separable covariate.

Fraction of units sorted is flat across recombination deciles (Fig. S2A). Standardised regression at the unit level gives a recombination coefficient of essentially zero ($\text{prop_fixed} \sim \text{recombination} + DI$: $\beta_{\text{recomb}} = 0.000$, $p = 0.97$) and a small but significant DI effect ($\beta_{DI} = -0.014$, $p < 10^{-16}$), consistent across recombination tertiles (Fig. S2C). Adding cluster size (the collider) spuriously induces a recombination effect, confirming it should be excluded. Thus **single-unit sorting is not concentrated in low-recombination regions** — consistent with the neutral expectation and in contrast with window-based reports, in which the low-recombination signal is partly the spatial-redundancy artefact that unit-level counting removes.

This is descriptive: "flat with recombination" is *consistent with* neutrality but does not by itself demonstrate absence of selection, which requires the recombination-matched null (S6). It also concerns single-locus sorting magnitude only; two-locus DMI signatures (correlated sorting of unlinked regions across populations) are tested separately.

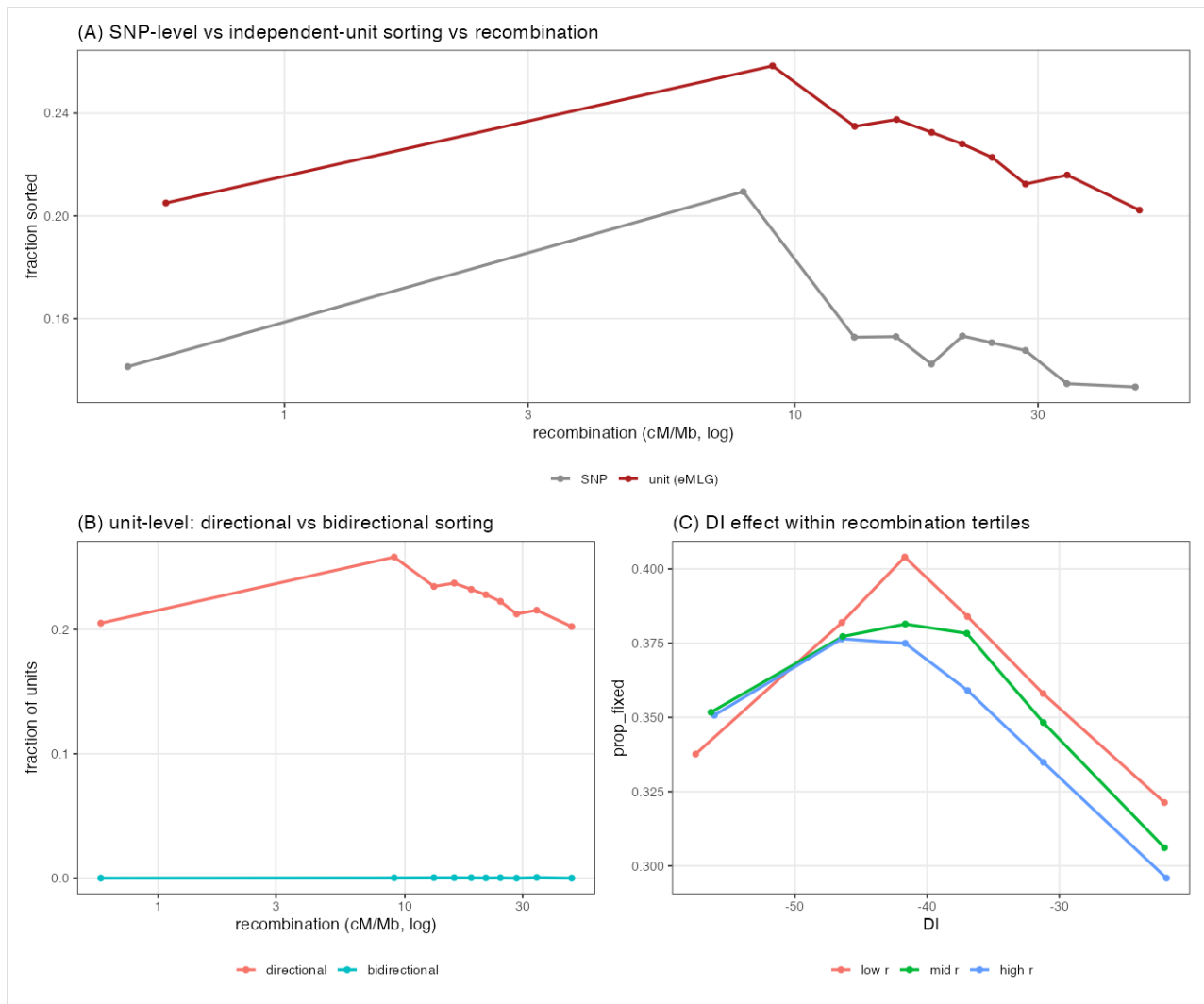


Figure S2. Sorting versus recombination. (A) Fraction sorted is flat at the independent-unit (eMLG) level; the SNP-level curve reflects spatial redundancy. (B) Unit-level sorting is almost entirely directional, with negligible bidirectional sorting. (C) The small DI effect is consistent across recombination tertiles, confirming separability.

S5. Interim conclusions

DI marks low-diversity, ancestry-informative sites spread across the recombination landscape rather than low-recombination regions (S3); single-unit sorting shows no recombination structure (S4); and sorting is overwhelmingly directional, with an aquilonia bias (S2). The descriptive framework thus isolates a predictable directional-sorting signal while showing that the obvious architectural confounders (recombination, DI, cluster size) are either orthogonal or controllable. Causal attribution to selection/DMIs is deferred to (i) a recombination-matched neutral null and (ii) an among-region (two-locus) co-sorting analysis.

S6. LD structure and neutral-null design

To test whether observed sorting exceeds drift, a haplodiploid forward simulation (SLiM) using the empirical recombination map is under construction. A key design decision is how to seed the parental founders. Seeding by independent per-SNP draws reproduces *admixture* LD (between-species allele-frequency differences) but not *within-species* haplotype LD. Composite- r^2 decay in matched panels ($n = 15$ diploids) shows within-species LD is minor where recombination is high (within/*admixture* $\approx$

15%) but large where recombination is low ($\approx 67\%$ overall, $\approx 85\%$ at short range; Fig. S3) — precisely the regions carrying the strongest sorting. This within-species LD is not removed by tightening the DI threshold. Founders will therefore be seeded from **real haploid-male haplotypes** (males are naturally phased in Hymenoptera), preserving within-species LD in the heritable (gametic) component; demographic parameters will be calibrated to empirical LD decay, heterozygosity and among-population F_{ST} , keeping the sorting statistics out of the calibration set.

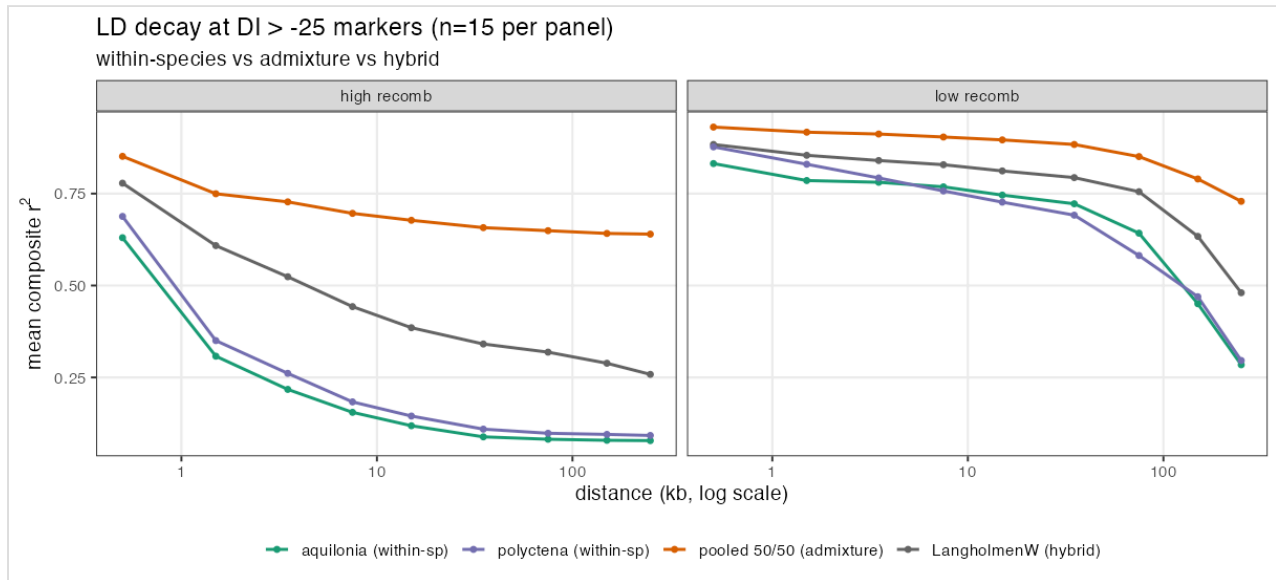


Figure S3. Composite- r^2 decay at diagnostic SNPs in matched panels. In high-recombination regions admixture (pooled-parent) LD dominates and within-species LD is minor; in low-recombination regions within-species LD is a large fraction of the admixture-scale LD — so independent-draw founder seeding would under-represent founder LD exactly where sorting is strongest.

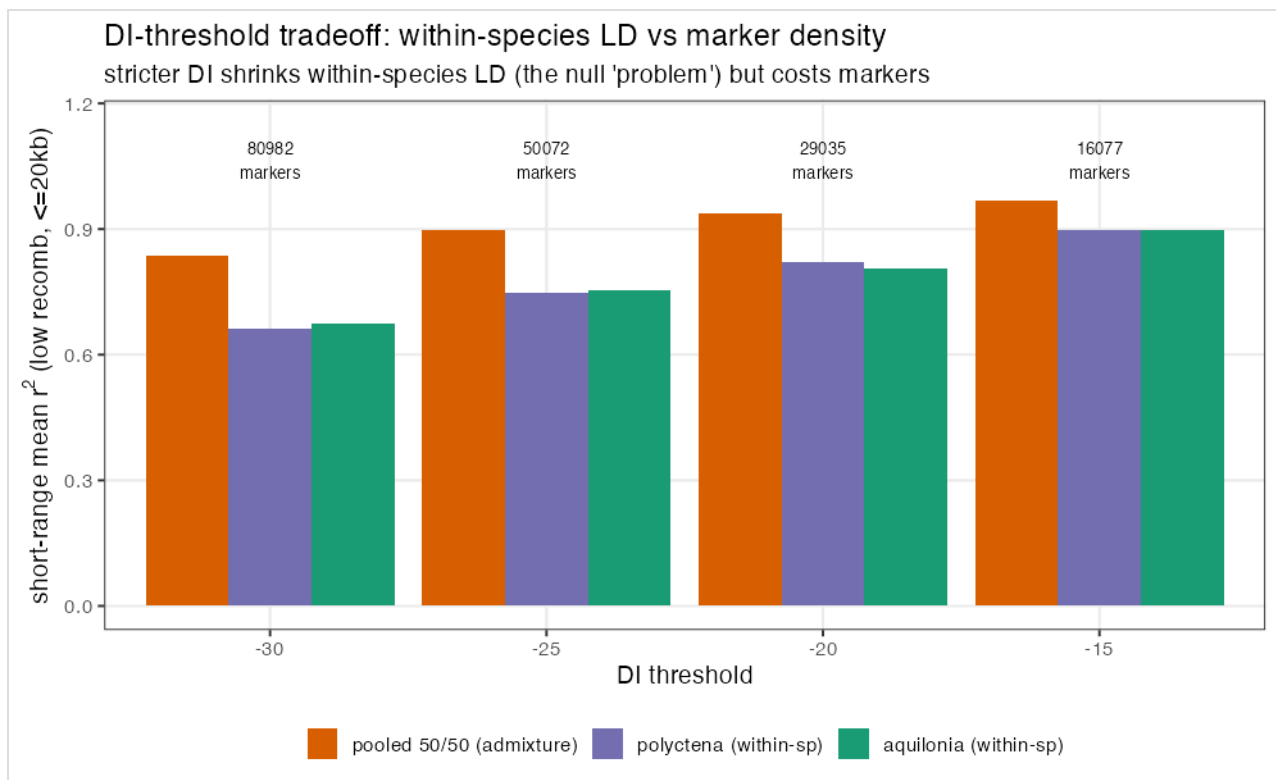


Figure S4. Within-species versus admixture short-range LD in low-recombination regions across DiagnosticIndex thresholds. The within-species fraction remains high (0.81–0.93) at every threshold, so the choice of DI cutoff cannot substitute for realistic haplotype seeding; marker-set and null-seeding decisions are therefore decoupled.

References

- Cruickshank T.E. & Hahn M.W. (2014) Reanalysis suggests that genomic islands of speciation are due to reduced diversity, not reduced gene flow. *Molecular Ecology* 23:3133–3157.
- Nouhaud P., Blanckaert A., Bank C. & Kulmuni J. (2022) Rapid and predictable genome evolution across three hybrid ant populations. *PLOS Biology* 20:e3001914.